ADITYA JHUNJHUNWALA

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SUMMARY

Aerospace Engineering student at IIEST Shibpur (**ISC 93.25% | ICSE 95.20%**) who built and shipped a **live multi-constellation GNSS health monitoring dashboard** — covering NavIC, QZSS, and BeiDou-3 Concurrently contributing to industry astrodynamics analysis and authoring monthly GNSS technical reports (*The Space PNT Report*) under professional mentorship. Skilled in Python-based orbital data processing, TLE propagation, and data visualisation for space domain applications.

EDUCATION

B.Tech in Aerospace Engineering — *2023 – 2027 (Expected)*
Indian Institute of Engineering Science and Technology (IIEST Shibpur), India — CGPA: 7.17 / 10 (Semesters 1–5)

Class XII — ISC — *2022*
St. Aloysius Orphanage and Day School, Howrah, India — 93.25%

Class X — ICSE — *2020*
St. Aloysius Orphanage and Day School, Howrah, India — 95.20%

EXPERIENCE

Space Domain Research Intern (Remote) — *Sep 2025 – Present*
Space Technology Consultant, Bengaluru

- Authored monthly GNSS technical reports (*The Space PNT Report*) covering 3+ navigation systems — analysing constellation health, orbital behaviour, and operational trends for industry distribution.
- Applied orbital mechanics and computational methods to model satellite system behaviour for space technology and commercial market assessments under direct industry mentorship.
- Developed Python scripts for TLE data processing, orbital propagation, and station-keeping analysis to support data-driven space domain research.
- Delivered astrodynamics-based insights informing commercial space project evaluations, translating raw orbital data into structured technical findings.

PROJECTS

GNSS Constellation Health Monitoring System — *2025 – 2026*
gnss-health-monitor.streamlit.app · GitHub Repo — Python · Streamlit · Space-Track API · Plotly · Folium

- Designed and deployed a real-time dashboard monitoring **30+ satellites** across NavIC, QZSS, and BeiDou-3, with automated TLE ingestion from the Space-Track API updating constellation state continuously.
- Engineered a custom weighted health-scoring algorithm combining orbital drift magnitude, station-keeping residuals, and inclination deviation into a single interpretable per-satellite health index.
- Built interactive multi-layer visualisations — time-series health trends, sky plots, and geographic ground-track maps — enabling simultaneous diagnostics across all 3 constellations.
- Implemented longitudinal slot deviation tracking with $\pm 180°$ wrap-around handling for accurate GEO slot occupancy monitoring across BeiDou-3's IGSO and GEO orbital regimes.

LEADERSHIP & EXTRACURRICULAR

Tech Coordinator — Abhiyantrix 2026 — *2026*
Departmental Tech Fest, IIEST Shibpur

- Built and deployed the official fest website (HTML / CSS / JS) for participant registrations, schedule display, and event information — live throughout the event.
- Coordinated logistics and execution of technical competitions and workshops, managing cross-team communication to ensure smooth event delivery.

TECHNICAL SKILLS

Programming: Python, C, SQL, HTML / CSS / JavaScript

Libraries & Tools: NumPy, Pandas, Matplotlib, Plotly, Folium, Streamlit, Git, GitHub, Space-Track API

Aerospace & Domain: Orbital Mechanics, Astrodynamics, GNSS & Satellite Navigation, TLE Analysis, SGP4 Propagation, Space Domain Awareness

Relevant Coursework: Aerospace Structures, Flight Dynamics, Propulsion, Thermodynamics, Engineering Mathematics

CERTIFICATIONS

CS50x: Introduction to Computer Science — Harvard University (edX) *2023*

- Completed problem sets in data structures, algorithms, and memory management across C, Python, SQL, and JavaScript.

Python Programming — Udemy *2022*

- Applied Python fundamentals — data structures, control flow, and functions — through mini-projects and practical exercises.